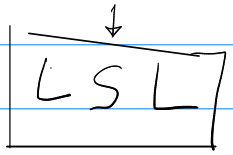
$2^1$

$(1011\,01)_2 \times 2 = (1011\,0110)_2$

$\times 4 = (1011\,0100)_2$

| | $b_7$ | $b_6$ | $b_5$ | $b_4$ | $b_3$ | $b_2$ | $b_1$ | $b_0$ | |
|---|---|---|---|---|---|---|---|---|---|
| 8 bit: | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | IN |
| | | | | | | | | | $\times 2$ |
| | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | OUT |

(LOGIC) SHIFT LEFT




| NUM_SHIFT N1 | NUM_SHIFT N0 | X2 | X4 | ~~X8~~ |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | |
| 0 | 1 | 1 | 0 | |
| 1 | 0 | 1 | 1 | |
| 1 | 1 | 0 | 0 | |

$X4 = N1 \cdot \overline{N0}$

$X2 = \overline{N_1} N0 + N1 \overline{N0} =$

$= N_1 \text{ XOR } N0$